Marcus Lin

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Education

University of California, Berkeley — Berkeley, CA
BA Data Science, BS Industrial Engineering & Operations Research — *August 2024 – May 2028*

Skills & Interests

Languages: Python, Java, Scala, SQL, TypeScript, JavaScript, HTML/CSS
Frameworks: SpringBoot, FastAPI, React, Node.js, Express, REST APIs
Developer Tools: Devin (Windsurf), Cursor, Claude Code, Codex, Git, Docker, AWS, Google Cloud Platform, Tableau
Libraries: scikit-learn, Hugging Face, XGBoost, pandas, NumPy, Matplotlib, Seaborn
Hobbies/Interests: Basketball, Scrolling, Food, Weightlifting, Football, Reading: Non-Fiction, Sci-Fi

Experience

Apache Spark — May 2026 – September 2026
Open Source Contributor — *Remote*

- Implemented Spark's variant_strip_nulls SQL function using Scala, Java, and Python, recursively removing null fields from VARIANT data across Catalyst, JVM binary-format logic, and Spark Connect APIs
- Extended Spark SQL's AVG aggregate to support the TIME type using Scala, implementing Catalyst division logic that preserves nanosecond precision
- Merged 5+ PRs collaborating with core maintainers on feature development, bug fixes, PySpark/Apache Arrow test coverage

NextEra Energy — May 2026 – August 2026
Data Science Intern — *Houston, TX*

- Built a production ML pipeline in Python using SQL, pandas, and XGBoost to predict customer churn within 90 days (81% ROC), loading model outputs to Microsoft SQL Server to enable targeted retention campaigns
- Optimized retention spend by 27% by automating an Excel profitability model into Python, enabling efficient batch processing of 5,000+ customers to determine which flagged at-risk customers were worth retaining
- Partnered with stakeholders to apply early churn model outputs and contracted margin analysis to retention strategy, prioritizing higher-margin customers and growing a lower winter flex risk customer book

InFLOWS AI — May 2025 – August 2025
Software Engineer Intern — *Berkeley, CA*

- Benchmarked open-source Hugging Face models for aspect-based sentiment analysis to replace GPT-4o in production-scale pipeline, enabling self-hosted inference and reducing per-inference costs by 95%
- Distilled GPT-4o outputs into Qwen2.5-3B via LoRA fine-tuning on a T4 GPU, achieving 3x inference throughput over the GPT-4o baseline while retaining 97% of its F1 score on aspect-based sentiment analysis
- Engineered a synthetic data generation pipeline creating 1000+ customer reviews for SFT, increasing model accuracy by 18%

Projects

Distributed Load Testing Platform | Go, gRPC, Docker, Kubernetes, AWS, Prometheus, Grafana | Source Code

- Orchestrated 10 containerized workers in Go via gRPC streaming to generate 75K+ concurrent requests against target services, validating autoscaling policies under sustained load
- Designed auto-scaling fleet management abstracted over Docker and EKS, dynamically provisioning workers based on real-time load demand and worker health status
- Instrumented end-to-end observability with Prometheus metrics and Grafana dashboards, capturing latency percentiles (p50/p95/p99), throughput, and error rates across distributed workers

NextEra Energy Hackathon | Python, React, Tailwind, FastAPI, Node.js | Source Code

- Developed a real-time EV charging platform using React, Tailwind, FastAPI, and Node.js to display availability across 106 campus chargers and recommend optimal charging times
- Built a recommendation engine that analyzes real-time session data to suggest available chargers and off-peak windows, distributing suggestions across time slots to prevent congestion and notification clustering

Leadership

Asia Consulting Enterprises — September 2026 – Present
Co-Founder — *Berkeley, CA*

- Co-founded a premier consulting group specializing in the Asian-Americas markets across technology, business, and fashion